

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		<i>Bordetella pertussis</i>	1			1		
		(ii)		Antigens , Lymphocytes Antibodies All 3 correct = 2 marks 2 correct = 1 mark	2			2		
	(b)			B			1	1		
	(c)	(i)		286.5/ 287(%) = 2 marks $7\,730 - 2\,000 / 2\,000 \times 100$ If incorrect award 1 mark for either Method correct with incorrect answer 286		2		2	2	
		(ii)		When vaccination rate {high/at target /95%} the number of cases was low. (1) When vaccination rate was {low/ decreased} number of cases {was high/ increased} (1)			2	2		
	(d)	(i)		(Whooping cough is) {serious /fatal/ can kill} (1) Vaccination rates are below { <u>target</u> percentage/ 95%} (1)			2	2		
		(ii)		Parental freedom of {choice /responsibility} / against religious beliefs/ (Worries) about side effects/ causes {harm/ allergies} Accept reference to concerns about autism e.g. they <u>think</u> it causes autism		1		1		
	(e)			(5) 1 (6) 2 (4) 3 All correct = 2 marks 2 correct = 1 mark		2		2		
				Total marks for question 4	3	5	5	13	2	0